

EC442: Lecture Notes

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Notes on the 1st year Macro class

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Contents

1. Lecture 1 Sep. 29	4
1.1. Why heterogeneity	4
1.2. How to model incomplete markets?	4
1.3. Planning problem	5
1.4. Towards a recursive formulation	6
1.5. Dynamic Programming	7
1.6. Contraction mapping theorem	7
1.6.1. Application to planner	8
1.7. Using the value function analytically	8
1.8. Using the value function numerically	8
2. Lecture 2 Oct 5	11
2.1. Recall	11
2.1.1. Bellman equation for the standard growth model	11
2.1.2. Contraction mapping theorem	11
2.2. Blackwell's sufficient conditions	11
2.2.1. Verification for the Bellman operator	11
2.3. Blackwell's conditions for the standard model	12
2.3.1. Monotonicity	12
2.3.2. Discounting	13
2.3.3. Boundedness and conclusion	13
2.4. Formulating recursive models	13
2.4.1. How to determine the state	14
2.4.2. Example: standard model with two sectors	14
2.4.3. Example: two sectors with irreversible investment	15
2.4.4. Standard model with shocks	16
2.4.4.1. States (k, z) : after the shock	16
2.4.4.2. State x : total resources	17
2.4.4.3. State k : before the shock	17
2.5. Equilibrium	18
2.6. What is a competitive equilibrium?	18
2.7. Sequential equilibrium	18
2.7.1. Household	18
2.7.2. Firm	19
2.7.3. Market clearing	19
2.7.4. No-Ponzi condition	20
2.8. Arrow-Debreu equilibrium	20
2.8.1. Household	21
2.8.2. Firm and market clearing	21
2.8.3. No arbitrage and the present-value budget	21
2.9. Recursive equilibrium	22
2.9.1. Standard model	22
2.10. The First Welfare Theorem	23
2.10.1. Key assumptions	23

2.10.2. Proof by contradiction of Theorem 2.10.1	23
2.10.3. Implication for the standard model	24
2.11. The Second Welfare Theorem	24
2.11.1. Sufficient conditions	24
2.11.2. Wealth transfers	25
2.11.3. Supporting prices	25
2.12. How do the welfare theorems matter?	25
3. Lecture 4 Oct 6	26
3.1. Fundamental models of economic growth	26
3.2. Solow	26
3.2.1. Setup	26
3.2.2. Capital per effective worker	27
3.2.3. Balanced growth path	27
3.2.4. Limitations	27
3.3. Malthus	28
3.3.1. Setup	28
3.3.2. Population per effective land unit	28
3.3.3. Balanced growth path	28
3.3.4. Malthus versus Solow	29
3.3.5. From Malthus to Solow	29
3.3.5.1. Technologies and resources	29
3.3.5.2. Households and population	30
3.3.5.3. Solow sector entry	31
3.3.5.4. Transition	31
3.3.5.5. Limitations	31
Bibliography	32

1. Lecture 1 Sep. 29

1.1. Why heterogeneity

Complete markets + standard welfare assumptions:

- Competitive eq. Pareto efficient; planner representation w/ fixed weights $\alpha_i > 0$.
- Common discounting; preferences $u(c) - v(n)$. Consumption FOCs:

$$\alpha_i u'(c_t^i) = \alpha_j u'(c_t^j) \Rightarrow \frac{u'(c_t^i)}{u'(c_t^j)} = \frac{\alpha_j}{\alpha_i}. \quad (1.1)$$

Marginal-utility ratio constant across time/states.

- Common CRRA: $u'(c) = c^{-\sigma}$.

$$\frac{c_t^i}{c_t^j} = \left(\frac{\alpha_i}{\alpha_j} \right)^{\frac{1}{\sigma}}. \quad (1.2)$$

Fixed consumption shares $\rightarrow$ **perfect insurance** against idiosyncratic income shocks. Aggregate risk remains; unequal Pareto weights $\rightarrow$ unequal consumption.

- Strictly convex labor disutility; interior labor choice; aggregate conditions fixed: productivity $\uparrow \rightarrow$ labor $\uparrow$; consumption share unchanged.

Implications:

- Karl Marx
- Full insurance at odds w/ data

Lecture addition: Intuition for Complete Markets

- **Example:** 18 students; one high earner, others lower. Complete markets $\rightarrow$ high earner sells state-contingent claims $\rightarrow$ consumption decoupled from income.
- **Marx analogy:** "From each according to ability, to each according to needs." Efficient allocation equalizes marginal utilities regardless of income.
- **Empirical mismatch:** Data shows strong income-consumption correlation. Complete markets imply correlation vanishes $\rightarrow$ model unrealistic. Heterogeneity requires market incompleteness.

Recording: 00:13:54, 00:16:04

1.2. How to model incomplete markets?

1. **Exogenous:** assume some markets absent.
 1. e.g. save/borrow via noncontingent bonds; no state-contingent claims.
2. **Endogenous:** derive market incompleteness from frictions.
 1. e.g. limited commitment (participation constraints), private information.
 2. deeper but harder to do

Lecture addition: Course Roadmap

- **Methods:** Dynamic programming, numerical methods.
- **Background:** Equilibrium and welfare theorems.
- **Growth:** Basic models of economic growth.
- **Incomplete markets:** Endogenous (participation constraints/limited commitment), exogenous (asset structure), search models.

Recording: 00:31:25, 00:33:02

Lecture addition: Frictions for Endogenous Incompleteness

- **Limited commitment:** Contracts unenforceable. E.g., US car insurance limits; agents flee jurisdiction (e.g., to Frankfurt) to avoid liability → limits risk trading.
- **Private information:** Effort/talent unobservable. Income sharing → moral hazard (effort reduction) → complete insurance inefficient/impossible.

Recording: 00:22:10, 00:27:38

1.3. Planning problem

Set $n_t = A = 1$; $f(k) = F(k, 1)$. Given $k_0 \geq 0$. Discount factor $0 < \beta < 1$; depreciation $0 \leq \delta \leq 1$.

Choose sequences $c_t, k_{t+1} \geq 0$:

$$\max \sum_{t=0}^{\infty} \beta^t u(c_t) \text{ s.t. } c_t + k_{t+1} = f(k_t) + (1 - \delta)k_t. \quad (1.3)$$

- **Lagrangian;** current-value multipliers λ_t :

$$\mathcal{L} = \sum_{t=0}^{\infty} \beta^t [u(c_t) + \lambda_t (f(k_t) + (1 - \delta)k_t - c_t - k_{t+1})]. \quad (1.4)$$

- **Interior FOCs:**

$$\begin{aligned} c_t : u'(c_t) &= \lambda_t, \\ k_{t+1} : \lambda_t &= \beta \lambda_{t+1} [f'(k_{t+1}) + 1 - \delta]. \end{aligned} \quad (1.5)$$

Euler equation:

$$u'(c_t) = \beta u'(c_{t+1}) [f'(k_{t+1}) + 1 - \delta]. \quad (1.6)$$

- **Transversality condition:**

$$\lim_{T \rightarrow \infty} \beta^T \lambda_T k_{T+1} = \lim_{T \rightarrow \infty} \beta^T u'(c_T) k_{T+1} = 0. \quad (1.7)$$

Discounted shadow value of terminal capital $\rightarrow 0$.

- Increasing, concave u ; concave f ; finite discounted utility: feasibility + FOCs + TVC $\rightarrow$ global optimum. At corners, use KKT conditions.

1.4. Towards a recursive formulation

- Sequence approach: infinitely many dated choices + optimality conditions.
- Repeated trade-off: consume today vs. save for tomorrow.
- Stationary problem; current capital k_t summarizes relevant history.
- Recursion: continuation value summarizes future choices; same decision problem at every date.

Value function: maximal lifetime utility from initial capital k_0 :

$$v^*(k_0) = \max_{\{c_t, k_{t+1}\}_{t \geq 0}} \sum_{t=0}^{\infty} \beta^t u(c_t). \quad (1.8)$$

Feasible sequences: $c_t, k_{t+1} \geq 0$; $c_t + k_{t+1} = f(k_t) + (1 - \delta)k_t$, all $t \geq 0$.

Split date 0 from continuation:

$$v^*(k_0) = \max_{c_0, k_1} \left[u(c_0) + \beta \max_{\{c_t, k_{t+1}\}_{t \geq 1}} \sum_{t=1}^{\infty} \beta^{t-1} u(c_t) \right]. \quad (1.9)$$

- Outer max: $c_0, k_1 \geq 0$; $c_0 + k_1 = f(k_0) + (1 - \delta)k_0$.
- Inner max: given chosen k_1 , same feasibility constraints for all $t \geq 1$. Discounting restarts at date 1 $\rightarrow$ inner value $v^*(k_1)$.

Continuation value absorbs future choices. **Bellman equation** (current capital k , next-period capital k'):

$$v^*(k) = \max_{c, k'} [u(c) + \beta v^*(k')] \quad (1.10)$$

subject to

$$c + k' = f(k) + (1 - \delta)k, c, k' \geq 0. \quad (1.11)$$

Infinite sequence $\rightarrow$ one-period choice + discounted continuation value. Stationarity $\rightarrow$ same problem at every date.

Bellman equation = **functional equation**; unknown = function $v(k)$.

Questions:

- **Existence:** conditions for solution v ?
- **Uniqueness:** conditions for unique v within chosen function class?
- **Equivalence:** when does $v = v^*$, with maximizing policy solving original sequence problem?

Answer:

- *Recursive Macroeconomic Theory* (RMT), “Dynamic Programming” ([Ljungqvist & Sargent, 2018, ch. 3](#)).

Lecture addition: Computational Advantage of DP

- **Lagrangian**: Leads to 2nd-order difference eq. Solving via “shooting” → local, clunky.
- **Recursive**: Bellman eq. → direct computation of **global solution** via iteration.
- **Convergence**: VFI guaranteed to converge to unique fixed point (via CMT), unlike many ODE/PDE methods.

Recording: 01:06:37, 01:07:39

1.5. Dynamic Programming

Bellman operator T : candidate continuation value $v \rightarrow$ updated value Tv .

$$(Tv)(k) = \max_{c,k'} [u(c) + \beta v(k')] \quad (1.12)$$

subject to

$$c + k' = f(k) + (1 - \delta)k, c, k' \geq 0. \quad (1.13)$$

Bellman solution = **fixed point**:

$$Tv = v \Leftrightarrow (Tv)(k) = v(k) \text{ for all feasible } k. \quad (1.14)$$

Fixed-point theorems give existence/uniqueness conditions within chosen function space; assumptions matter.

1.6. Contraction mapping theorem

Theorem 1.6.1 (Contraction mapping theorem): Let (S, d) be a nonempty complete metric space and $T : S \rightarrow S$ a contraction: there exists $\beta \in (0, 1)$ such that

$$d(Tx, Ty) \leq \beta d(x, y) \text{ for all } x, y \in S. \quad (1.15)$$

Then T has a unique fixed point $v \in S$. For every $v_0 \in S$ and $n = 0, 1, 2, \dots$,

$$d(T^n v_0, v) \leq \beta^n d(v_0, v). \quad (1.16)$$

Hence $T^n v_0 \rightarrow v$ as $n \rightarrow \infty$, with geometric convergence. Here T^n denotes n successive applications of T , with T^0 the identity.

Lecture addition: Intuition for Contraction Mapping

- Analogous to Brouwer fixed-point theorem in 1D: continuous function mapping interval to itself with slope $< 1 \rightarrow$ unique fixed point.
- In function spaces, “slope” = contraction factor β . Iterating operator moves points closer $\rightarrow$ convergence to unique fixed point.

Recording: 00:59:06, 01:01:46

1.6.1. Application to planner

- Assume feasible capital remains in nonempty compact interval K .
- $S = C(K)$: continuous real-valued functions on K ; automatically bounded.
- **Sup-norm metric**:

$$d(v, w) = \|v - w\|_\infty = \sup_{k \in K} |v(k) - w(k)|. \quad (1.17)$$

(S, d) complete.

- Must verify $T : S \rightarrow S$: updated values remain bounded and continuous. Then Bellman discounting gives

$$\|Tv - Tw\|_\infty \leq \beta \|v - w\|_\infty. \quad (1.18)$$

Theorem 1.6.1 $\rightarrow$ unique fixed point in S ; value iteration converges from any $v_0 \in S$.

Lecture addition: Economic Interpretation of VFI

- **Initial guess** $v^0 = 0$: One-period model (no future utility). Agent consumes all output today.
- **First iteration** Tv^0 : Two-period model. Optimize today/tomorrow, world ends after tomorrow.
- **Convergence**: Each iteration extends horizon by 1 period. $n \rightarrow \infty \rightarrow$ infinite-horizon value function.

Recording: 00:56:48, 00:58:31

1.7. Using the value function analytically

Assume u , f , and v differentiable; optimal choices interior. Substitute $c = f(k) + (1 - \delta)k - k'$ into Bellman:

$$v(k) = \max_{k'} [u(f(k) + (1 - \delta)k - k') + \beta v(k')], \quad (1.19)$$

over feasible k' .

- **FOC for k'** :

$$-u'(c) + \beta v'(k') = 0 \Rightarrow u'(c) = \beta v'(k'). \quad (1.20)$$

- **Envelope condition**: differentiate w.r.t. k , holding optimal k' fixed:

$$v'(k) = u'(c)[f'(k) + 1 - \delta]. \quad (1.21)$$

- **Combine**: apply envelope condition next period $\rightarrow$ usual Euler equation:

$$u'(c_t) = \beta v'(k_{t+1}) = \beta u'(c_{t+1})[f'(k_{t+1}) + 1 - \delta]. \quad (1.22)$$

Same condition as sequence problem. At corners, use KKT conditions.

1.8. Using the value function numerically

Need numerical representation of $v(k)$.

- Simplest approach: restrict capital to finite grid

$$K_N = \{k^1, k^2, \dots, k^N\} \subset K. \quad (1.23)$$

- Approximate value function by vector

$$\mathbf{v} = (v_1, v_2, \dots, v_N)^\top \in \mathbb{R}^N, v_i \approx v(k^i). \quad (1.24)$$

One entry per capital grid point.

Algorithm 1.8.1 (Value function iteration (VFI)):

1. **Choose grid** $K_N = \{k^1, \dots, k^N\}$. For each pair (i, j) , compute

$$c_{ij} = f(k^i) + (1 - \delta)k^i - k^j. \quad (1.25)$$

Let J_i contain choices j with $c_{ij} \geq 0$ and finite $u(c_{ij})$. Each J_i must be nonempty.

2. **Initial guess**: any $\mathbf{v}^0 \in \mathbb{R}^N$, e.g. all zeros. Set iteration $m = 0$ and tolerance $\varepsilon > 0$.
3. **Bellman update**: for every $i = 1, \dots, N$,

$$v_i^{m+1} = \max_{j \in J_i} [u(c_{ij}) + \beta v_j^m]. \quad (1.26)$$

Use the old vector for all updates: $\mathbf{v}^{m+1} = T_N \mathbf{v}^m$.

4. **Convergence check**:

$$\Delta_m = \max_{1 \leq i \leq N} |v_i^{m+1} - v_i^m|. \quad (1.27)$$

If $\Delta_m \leq \varepsilon$, stop and return $\hat{\mathbf{v}} = \mathbf{v}^{m+1}$. Otherwise set $m \leftarrow m + 1$ and repeat step 3.

5. **Recover approximate policy**: for each i , choose

$$j_i \in \arg \max_{j \in J_i} [u(c_{ij}) + \beta \hat{v}_j]. \quad (1.28)$$

Set $k'(k^i) = k^{j_i}$ and $c(k^i) = c_{ij_i}$.

Why it converges: T_N is a β -contraction in the sup norm. **Theorem 1.6.1** $\rightarrow$ unique grid fixed point $\mathbf{v}_N^*$; **Algorithm 1.8.1** converges from any initial vector. At stopping, value-error bound:

$$\|\hat{\mathbf{v}} - \mathbf{v}_N^*\|_\infty \leq \frac{\beta}{1 - \beta} \Delta_m \leq \frac{\beta}{1 - \beta} \varepsilon. \quad (1.29)$$

This controls iteration error; grid approximation error remains.

Lecture addition: Convergence in General Equilibrium

- CMT guarantees convergence for individual decision problems (value functions).
- Heterogeneous GE: convergence of **distribution** of states (assets, income, human capital) required.
- Distribution = higher-dimensional object; may have multiple fixed points → more complex than single-agent VFI.

Recording: 01:14:16, 01:15:18

Lecture addition: Curse of Dimensionality

- Grid-based VFI suffers from **curse of dimensionality**: n state variables → M^n grid points (exponential growth).
- Large n (e.g., 50) → computationally intractable.
- **Frontier approaches**: Sparse grids; mean field games [ASR uncertain: Ben Moll]; heuristic trust in convergence when rigorous proofs unavailable.

Recording: 01:15:51, 01:16:53

Lecture addition: Complexity vs. Reality

- Model complexity driven by **expectation side**: agents anticipate others' actions → complex fixed-point problems.
- Evidence: real people not “fully forward-looking” as standard models assume.
- **Responses**: (1) Develop computational tools for complex models; (2) Simplify models to match cognitive complexity (behavioral/limited rationality).

Recording: 01:17:25, 01:19:00

2. Lecture 2 Oct 5

2.1. Recall

2.1.1. Bellman equation for the standard growth model

$$v(k) = \max_{c,k'} [u(c) + \beta v(k')], u(c) = \frac{c^{1-\sigma}}{1-\sigma}, \quad (2.1)$$

subject to

$$c + k' = (1 - \delta)k + k^\alpha, c > 0, k' \geq 0. \quad (2.2)$$

For $\sigma = 1$, use $u(c) = \log c$.

2.1.2. Contraction mapping theorem

On a nonempty complete metric space (S, d) , a contraction $T : S \rightarrow S$ with modulus $\beta \in (0, 1)$ has a unique fixed point v (**Theorem 1.6.1**):

$$\begin{aligned} d(Tx, Ty) &\leq \beta d(x, y), \\ d(T^n v_0, v) &\leq \beta^n d(v_0, v). \end{aligned} \quad (2.3)$$

Iteration from any $v_0 \in S$ converges geometrically to v .

2.2. Blackwell's sufficient conditions

Theorem 2.2.1 (Blackwell's sufficient conditions): Let $B(X)$ be the bounded real-valued functions on X , equipped with the sup norm, and let $T : B(X) \rightarrow B(X)$. Suppose:

1. **Monotonicity**: For all $f, g \in B(X)$,

$$f(x) \geq g(x) \text{ for all } x \in X \implies (Tf)(x) \geq (Tg)(x) \text{ for all } x \in X. \quad (2.4)$$

2. **Discounting**: There exists $\beta \in (0, 1)$ such that, for every $f \in B(X)$ and $a \geq 0$,

$$(T(f + a))(x) \leq (Tf)(x) + \beta a \text{ for all } x \in X. \quad (2.5)$$

Here $(f + a)(x) = f(x) + a$.

Then T is a contraction with modulus β :

$$\|Tf - Tg\|_\infty \leq \beta \|f - g\|_\infty. \quad (2.6)$$

2.2.1. Verification for the Bellman operator

Maximization preserves pointwise inequalities, so $f \geq g \implies Tf \geq Tg$. Adding a constant to continuation utility gives

$$(T(f + a))(k) = \max_{c,k'} [u(c) + \beta f(k')] + \beta a = (Tf)(k) + \beta a. \quad (2.7)$$

Both conditions hold; the self-map requirement $T : B(X) \rightarrow B(X)$ must also be verified.

2.3. Blackwell's conditions for the standard model

Let $\alpha, \delta, \beta \in (0, 1)$. Resources and consumption:

$$y(k) = \underbrace{(1 - \delta)k}_{\text{remaining capital}} + \underbrace{k^\alpha}_{\text{output}}, \quad (2.8)$$

$$c(k, k') = y(k) - k'.$$

The invariant state space satisfies

$$\bar{k} = \delta^{-\frac{1}{1-\alpha}},$$

$$X = [0, \bar{k}], \quad (2.9)$$

$$0 \leq y(k) \leq y(\bar{k}) = \bar{k} \text{ for all } k \in X.$$

Lecture addition: The capital ceiling

$\bar{k}$ is the steady state with zero consumption: all output goes into investment, and $\bar{k}^\alpha = \delta\bar{k}$. The bound comes from feasibility alone. Starting in X , feasible savings remain in X .

Recording: 00:07:02 - 00:07:39

Feasible savings and the Bellman operator:

$$\Gamma(k) = \{k' \geq 0 : c(k, k') \geq 0\} \subseteq X,$$

$$(Tf)(k) = \sup_{k' \in \Gamma(k)} \left[\underbrace{u(c(k, k'))}_{\text{current utility}} + \beta \underbrace{f(k')}_{\text{continuation value}} \right]. \quad (2.10)$$

Extend u to $c = 0$ by its limit, possibly $-\infty$. Replace $\sup$ with $\max$ when the optimum is attained.

2.3.1. Monotonicity

For $f, g \in B(X)$ with $f \geq g$ pointwise,

$$\begin{aligned} (Tf)(k) &= \sup_{k' \in \Gamma(k)} [u(c(k, k')) + \beta f(k')] \\ &\geq \sup_{k' \in \Gamma(k)} [u(c(k, k')) + \beta g(k')] \\ &= (Tg)(k). \end{aligned} \quad (2.11)$$

Lecture addition: Why the first inequality can be strict

If the maximum for g is attained, let $k^*(k)$ be its optimal savings:

$$\begin{aligned}(Tf)(k) &\geq u(c(k, k^*(k))) + \beta f(k^*(k)), \\ &\geq u(c(k, k^*(k))) + \beta g(k^*(k)), \\ &= (Tg)(k).\end{aligned}$$

The first step evaluates one feasible choice. It can be strict because $k^*(k)$ need not maximize the objective with continuation value f .

Recording: 00:08:00 - 00:10:59

2.3.2. Discounting

For $f \in B(X)$ and a constant $a \geq 0$,

$$\begin{aligned}(T(f+a))(k) &= \sup_{k' \in \Gamma(k)} \left[u(c(k, k')) + \beta f(k') + \underbrace{\beta a}_{\text{constant in } k'} \right] \\ &= (Tf)(k) + \beta a.\end{aligned}\tag{2.12}$$

2.3.3. Boundedness and conclusion

For $0 < \sigma < 1$, $u(0) = 0$ and

$$\begin{aligned}0 \leq u(c(k, k')) &\leq u(\bar{k}) < \infty, \\ \|Tf\|_\infty &\leq u(\bar{k}) + \beta \|f\|_\infty < \infty.\end{aligned}\tag{2.13}$$

Thus $T : B(X) \rightarrow B(X)$ on the complete sup-norm space $B(X)$. By [Theorem 2.2.1](#) and [Theorem 1.6.1](#), there is a unique $v \in B(X)$ with $Tv = v$, and for every $v_0 \in B(X)$,

$$\|T^n v_0 - v\|_\infty \leq \beta^n \|v_0 - v\|_\infty \rightarrow 0.\tag{2.14}$$

For $\sigma \geq 1$ (including log utility),

$$k = 0 \implies \Gamma(k) = \{0\} \implies (Tf)(0) = -\infty.\tag{2.15}$$

The bounded-function theorem then requires a different state or function space.

2.4. Formulating recursive models

Identify the state variables before writing the Bellman equation.

A **state variable**:

- Matters for the decision problem.
- Is taken as given by the decision maker at the time of choice.
- Varies over time or across states.

Other variables:

- **Choice variables**: chosen by the decision maker.

- **Parameters:** affect the problem but are fixed over time and across states.

The value function takes the state variables as its arguments.

Lecture addition: A past choice can be a current state

The household chose today's capital in the past; it is fixed when today's consumption-saving decision is made. Models with habits or varying patience may require preference states. Here β and σ remain parameters because they are fixed.

Recording: 00:16:22 - 00:18:54

2.4.1. How to determine the state

1. **Fix timing:** Specify what is observed before choices are made. A lagged choice may be today's state; a newly chosen quantity is a control.
2. **List candidates:** Retain inherited stocks and observed shocks that affect current constraints, utility, or forecasts. Separate fixed parameters.
3. **Check sufficiency:** Given state s and choices q , specify

$$\begin{aligned} q &\in \Gamma(s), \\ \text{current utility} &= U(s, q), \\ s' &= G(s, q, \varepsilon'). \end{aligned} \tag{2.16}$$

Here ε' is future uncertainty. Two histories with the same s must give the same feasible choices, current utility, and conditional distribution of s' for every feasible q .

4. **Remove redundancy:** Drop a coordinate only if this sufficiency test still holds. A one-to-one change of coordinates gives an equivalent state.
5. **Write the Bellman equation:**

$$v(s) = \sup_{q \in \Gamma(s)} \left[\underbrace{U(s, q)}_{\text{current utility}} + \beta \underbrace{\mathbb{E}[v(s') | s, q]}_{\text{expected continuation}} \right]. \tag{2.17}$$

Use max when the optimum is attained; omit the expectation in a deterministic model.

Reduction test: If two candidate states have the same proposed summary but different feasible choices, utility, or transition distributions, the summary is insufficient.

2.4.2. Example: standard model with two sectors

Each period, allocate inherited capital k and one unit of labor freely between consumption (c) and investment (i) sectors. Let $\alpha, \gamma \in (0, 1)$.

Feasibility:

$$\begin{aligned}
 c &= \underbrace{k_c^\alpha n_c^{1-\alpha}}_{\text{consumption output}}, \\
 k' &= \underbrace{(1-\delta)k}_{\text{remaining capital}} + \underbrace{k_i^\gamma n_i^{1-\gamma}}_{\text{investment output}}, \\
 k_c + k_i &= k, \\
 n_c + n_i &= 1.
 \end{aligned} \tag{2.18}$$

All quantities are nonnegative.

State identification:

- k is inherited; (k_c, k_i, n_c, n_i) are chosen before production.
- Free reallocation removes dependence on the previous sectoral split.
- Given k and current choices, utility and next-period k' are determined.

Hence $s = k$ is sufficient.

The Bellman equation is

$$v(k) = \max_{c, k', k_c, k_i, n_c, n_i} [u(c) + \beta v(k')], \tag{2.19}$$

subject to the constraints above, with u as defined in Recall.

Lecture addition: Timing and the sectoral split

Let $\tilde{v}(k_c, k_i)$ be the value after capital has been allocated across sectors. Before allocation,

$$v(k) = \max_{k_c + k_i = k} \tilde{v}(k_c, k_i).$$

In this deterministic model, the future split can be planned a period earlier. Treating the initial allocation consistently gives the same choices. With M grid points per state, retaining both stocks requires M^2 combinations; retaining total capital requires M .

Recording: 00:24:30 - 00:30:44

2.4.3. Example: two sectors with irreversible investment

Capital is installed before the period and cannot move between sectors. Choose labor allocations and new investment x_c, x_i today; investment becomes productive next period.

Feasibility:

$$\begin{aligned}
c &= k_c^\alpha n_c^{1-\alpha}, \\
x_c + x_i &= \underbrace{k_i^\gamma n_i^{1-\gamma}}_{\text{investment output}}, \\
k'_c &= (1 - \delta)k_c + x_c, \\
k'_i &= (1 - \delta)k_i + x_i, \\
n_c + n_i &= 1, \\
x_c, x_i &\geq 0.
\end{aligned} \tag{2.20}$$

All other quantities are nonnegative. Irreversibility implies

$$k'_j \geq (1 - \delta)k_j \text{ for } j \in \{c, i\}. \tag{2.21}$$

Sectoral capital can decline only through depreciation.

Lecture addition: Why the aggregate equation is insufficient

The state is (k_c, k_i) . Adding the accumulation equations leaves the separate restrictions $x_c, x_i \geq 0$ in place. Dropping them would permit conversion of existing capital between sectors. They can bind when initial stocks are imbalanced or shocks change the desired sectoral mix.

Recording: 00:32:20 - 00:34:45

Lecture addition: Applications of multisector models

Separate consumption and investment sectors allow general and investment-specific technological change to have different effects on business cycles. Larger models use input-output networks, with sectors buying CES aggregates of intermediate inputs, to study shock propagation.

Recording: 00:20:42 - 00:20:53; 00:35:00 - 00:36:15

2.4.4. Standard model with shocks

Productivity z is i.i.d. over time. The resource constraint is

$$c + k' = (1 - \delta)k + zk^\alpha. \tag{2.22}$$

In all formulations, $c, k' \geq 0$.

2.4.4.1. States (k, z) : after the shock

Observe z , then choose c and k' :

$$v(k, z) = \max_{c, k'} [u(c) + \beta \mathbb{E}_{z'} [v(k', z')]], \tag{2.23}$$

subject to [Equation \(2.22\)](#). The expectation is over next period's shock z' .

2.4.4.2. State x : total resources

After observing z , define

$$x = (1 - \delta)k + zk^\alpha. \quad (2.24)$$

Then

$$w(x) = \max_{c, k'} [u(c) + \beta \mathbb{E}_{z'} [w(x')]], \quad (2.25)$$

subject to

$$\begin{aligned} c + k' &= x, \\ x' &= (1 - \delta)k' + z'(k')^\alpha. \end{aligned} \quad (2.26)$$

z' is independent of z , so the transition of x' requires only k' .

Note: This reduction to x relies on z being i.i.d. over time.

Lecture addition: The forecast matters with persistent shocks

Two states can have the same current resources x but different productivity z . With persistence, their distributions of z' differ, so their optimal savings can differ. Current z carries information about tomorrow.

Recording: 00:45:42 - 00:46:22

2.4.4.3. State k : before the shock

$V(k)$ is the value *before* z is observed. Choices $c(z)$ and $k'(z)$ may depend on its realization:

Choices. For a fixed (k, z) or x , Options 1 and 2 maximize over scalar choices c, k' . Their policy functions are $c(k, z), k'(k, z)$ or $c(x), k'(x)$; state arguments are implicit. Option 3 maximizes over functions $c(z), k'(z)$ for every possible z .

$$V(k) = \max_{c(z), k'(z)} \mathbb{E}_z [u(c(z)) + \beta V(k'(z))], \quad (2.27)$$

subject to

$$c(z) + k'(z) = (1 - \delta)k + zk^\alpha \text{ for every } z. \quad (2.28)$$

$$\begin{aligned} v(k, z) &= w((1 - \delta)k + zk^\alpha), \\ V(k) &= \mathbb{E}_z [v(k, z)]. \end{aligned} \quad (2.29)$$

Timing. Before observing z , $V(k)$ averages over its realization; i.i.d. shocks make past productivity uninformative about z . After observing z , the state records realized resources through (k, z) or x . Decisions can depend on observed z in all three formulations.

Lecture addition: Several values within one period

A model can use value functions at several stages of a period. A value before information arrives is linked to the later value by an expectation over the information revealed between those stages.

Recording: 00:51:25 - 00:52:16

2.5. Equilibrium

A social planner maximizes welfare subject to feasibility constraints. In a decentralized economy, agents make their own choices, linked through markets, institutions, and policies.

- How can decentralized outcomes be represented formally?
- How do equilibrium allocations relate to the planner's solution?
- Can decentralized equilibria be formulated recursively?

2.6. What is a competitive equilibrium?

A competitive equilibrium consists of:

- An *allocation*: consumption, production, and factor inputs.
- A *price system*: prices in every market.

Two conditions:

- **Optimization**: Households maximize utility and firms maximize profits, taking prices as given.
- **Feasibility**: The allocation is feasible: total use of each good does not exceed production plus endowments.

Lecture addition: Trading dates and events

With complete markets, goods are distinguished by delivery date and event. The lecture's example is consumption in 2027 conditional on no rain in Australia. Suitable sequential asset markets can span the same contingent consumption trades as markets opened once at date 0.

Recording: 01:02:26 - 01:03:56

2.7. Sequential equilibrium

In the deterministic standard model, markets open each period. The household owns capital k_t and a fixed labor endowment $n_t = 1$. It rents capital and supplies labor to the firm; both agents take prices as given. Let r_t be the net return on capital. The capital rental rate is $r_t + \delta$; the wage is w_t .

Given k_0 , a sequential equilibrium consists of an allocation $\{c_t, n_t, k_{t+1}\}_{t=0}^{\infty}$ and prices $\{r_t, w_t\}_{t=0}^{\infty}$ satisfying the following conditions.

2.7.1. Household

Given the price paths, the household solves

$$\max_{\{c_t, k_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(c_t), \quad (2.30)$$

subject to

$$\begin{aligned} c_t + k_{t+1} &= \underbrace{(1 + r_t)k_t}_{\text{capital payoff}} + \underbrace{w_t n_t}_{\text{labor income}}, \\ n_t &= 1, \\ c_t &\geq 0, \\ k_{t+1} &\geq 0. \end{aligned} \quad (2.31)$$

2.7.2. Firm

The firm chooses capital demand K_t and labor demand N_t each period:

$$\max_{K_t, N_t \geq 0} [K_t^\alpha N_t^{1-\alpha} - (r_t + \delta)K_t - w_t N_t]. \quad (2.32)$$

Lecture addition: Representative agents and price taking

The representative firm behaves as an atomistic price taker. Its labor demand N_t is a choice; $N_t = 1$ is imposed through market clearing. Representative notation can stand for a continuum of identical agents. Separate demand and supply symbols make this distinction explicit.

Recording: 01:06:36 - 01:08:06; 01:12:24 - 01:13:53

2.7.3. Market clearing

$$\begin{aligned} K_t &= k_t, \\ N_t &= n_t = 1, \\ c_t + k_{t+1} &= (1 - \delta)k_t + k_t^\alpha. \end{aligned} \quad (2.33)$$

At positive inputs, the firm's first-order conditions imply

$$\begin{aligned} r_t &= \alpha k_t^{\alpha-1} - \delta, \\ w_t &= (1 - \alpha)k_t^\alpha. \end{aligned} \quad (2.34)$$

Lecture addition: Depreciation and the quoted return

An alternative convention quotes the capital rental rate $\rho_t = r_t + \delta$ and lets the household bear depreciation. Then

$$\begin{aligned} c_t + k_{t+1} &= (1 - \delta + \rho_t)k_t + w_t n_t, \\ \text{firm capital cost} &= \rho_t K_t. \end{aligned}$$

The net-return and rental-rate conventions describe the same payments.

Recording: 01:10:41 - 01:11:28; 01:14:52 - 01:16:21

2.7.4. No-Ponzi condition

If borrowing is allowed, replace $k_{t+1} \geq 0$ with

$$\liminf_{t \rightarrow \infty} \frac{k_{t+1}}{\prod_{s=1}^t (1 + r_s)} \geq 0. \quad (2.35)$$

At an interior optimum with $u'(c_s) > 0$, the Euler equation gives

$$1 + r_s = \frac{u'(c_{s-1})}{\beta u'(c_s)}. \quad (2.36)$$

Multiplying from $s = 1$ to t , marginal utilities cancel:

$$\begin{aligned} \prod_{s=1}^t (1 + r_s) &= \prod_{s=1}^t \frac{u'(c_{s-1})}{\beta u'(c_s)}, \\ &= \frac{u'(c_0)}{\beta^t u'(c_t)}. \end{aligned} \quad (2.37)$$

Hence

$$\frac{k_{t+1}}{\prod_{s=1}^t (1 + r_s)} = \frac{\beta^t u'(c_t) k_{t+1}}{u'(c_0)}. \quad (2.38)$$

Since $u'(c_0) > 0$ is constant, dividing by it preserves the sign of the limit inferior. Thus the no-Ponzi condition is equivalent to

$$\liminf_{t \rightarrow \infty} \beta^t u'(c_t) k_{t+1} \geq 0. \quad (2.39)$$

The household cannot finance consumption by rolling over debt forever.

Lecture addition: Physical capital and financial assets

Physical capital is nonnegative. An individual's financial asset position can be negative when borrowing is allowed for consumption smoothing. Period budgets alone permit arbitrarily large debt followed by refinancing of principal and interest. A borrowing limit or no-Ponzi condition restricts those plans.

Recording: 01:16:28 - 01:17:20; 01:18:29 - 01:19:47

2.8. Arrow-Debreu equilibrium

Sequential markets open each period. In an Arrow-Debreu equilibrium, markets open once at date 0. Agents contract for goods delivered at each future date (and state, under uncertainty). Delivery and production still occur each period.

Given k_0 , an Arrow-Debreu equilibrium consists of an allocation $\{c_t, n_t, k_{t+1}\}_{t=0}^{\infty}$ and prices $\{p_t, r_t, w_t\}_{t=0}^{\infty}$ satisfying the conditions below.

The additional price $p_t > 0$ is the date-0 price of one unit of the good delivered at date t , with $p_0 = 1$. It converts date- t quantities into date-0 goods.

The wage w_t and capital rental rate $r_t + \delta$ remain measured in date- t goods. Their date-0 prices are $p_t w_t$ and $p_t(r_t + \delta)$.

2.8.1. Household

The household solves

$$\max_{\{c_t, k_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(c_t), \quad (2.40)$$

subject to $n_t = 1$, $c_t \geq 0$, given k_0 , and a single present-value budget:

$$\sum_{t=0}^{\infty} p_t(c_t + k_{t+1}) = \sum_{t=0}^{\infty} p_t[(1 + r_t)k_t + w_t n_t]. \quad (2.41)$$

The present-value sums are assumed finite.

2.8.2. Firm and market clearing

For each delivery date t , the firm solves

$$\max_{K_t, N_t \geq 0} [K_t^\alpha N_t^{1-\alpha} - (r_t + \delta)K_t - w_t N_t]. \quad (2.42)$$

Multiplying profits by $p_t > 0$ leaves the input choices unchanged. Markets clear at every date:

$$\begin{aligned} K_t &= k_t, \\ N_t &= n_t = 1, \\ c_t + k_{t+1} &= (1 - \delta)k_t + k_t^\alpha. \end{aligned} \quad (2.43)$$

2.8.3. No arbitrage and the present-value budget

Definition 2.8.1 (No arbitrage in capital): In this deterministic Arrow-Debreu economy, one unit of k_{t+1} costs p_t at date 0 and pays $p_{t+1}(1 + r_{t+1})$ in date-0 goods. No arbitrage in capital requires

$$\begin{aligned} p_t - p_{t+1}(1 + r_{t+1}) &= 0, \\ \frac{p_t}{p_{t+1}} &= 1 + r_{t+1}. \end{aligned} \quad (2.44)$$

Thus $\frac{p_t}{p_{t+1}}$ is the gross real return from t to $t + 1$. With $p_0 = 1$,

$$p_t = \frac{1}{\prod_{s=1}^t (1 + r_s)}. \quad (2.45)$$

Capital purchases and their future payoffs cancel from the lifetime budget:

$$\sum_{t=0}^{\infty} p_t c_t = \underbrace{(1+r_0)k_0}_{\text{initial capital wealth}} + \underbrace{\sum_{t=0}^{\infty} p_t w_t}_{\text{lifetime labor income}}. \quad (2.46)$$

The lifetime budget makes a separate no-Ponzi condition unnecessary.

2.9. Recursive equilibrium

Recursive methods apply to **decentralized economies** as well as planning problems. Build on **sequential equilibrium**: agents choose each period, and state variables evolve over time.

In the standard model, distinguish two states:

- k : **individual capital**, which determines household wealth.
- K : **aggregate capital**, which determines prices and their future path.

The household chooses k' taking $K' = g(K)$ and the associated prices *as given*. In equilibrium, $k = K$, but keeping both states in the household problem preserves **price-taking**. With heterogeneity, individual and aggregate states generally differ.

2.9.1. Standard model

A recursive competitive equilibrium consists of:

- **Value function**: $v(k, K)$.
- **Policy functions**: $c(k, K)$ and $k'(k, K)$.
- **Price functions**: $r(K)$ and $w(K)$.
- **Aggregate law of motion**: $g(K)$.

These functions satisfy four conditions:

1. **Household optimality**. Given r , w and g , the value function solves

$$v(k, K) = \max_{c, k' \geq 0} \{u(c) + \beta v(k', K')\}, \quad (2.47)$$

subject to

$$\begin{aligned} c + k' &= (1 + r(K))k + w(K), \\ K' &= g(K). \end{aligned} \quad (2.48)$$

The policies $c(k, K)$ and $k'(k, K)$ attain this maximum.

2. **Firm optimality**. With aggregate labor equal to one and $K > 0$,

$$\begin{aligned} r(K) &= \alpha K^{\alpha-1} - \delta, \\ w(K) &= (1 - \alpha)K^{\alpha}. \end{aligned} \quad (2.49)$$

Here $r(K)$ is the *net return*; the capital rental rate is $r(K) + \delta$.

3. **Market clearing**. For every admissible K ,

$$c(K, K) + k'(K, K) = (1 - \delta)K + K^{\alpha}. \quad (2.50)$$

4. **Consistency.** The *perceived* aggregate law must equal the law *generated* by optimal household saving:

$$g(K) = k'(K, K). \quad (2.51)$$

2.10. The First Welfare Theorem

Theorem 2.10.1 (First Welfare Theorem): With **locally nonsatiated preferences**, **complete competitive markets** and **no externalities**, every competitive equilibrium allocation is **Pareto efficient**. This formulation assumes finitely many consumers and well-defined lifetime budgets.

Pareto efficiency means that no feasible allocation makes everyone weakly better off and at least one person strictly better off.

2.10.1. Key assumptions

- **Local nonsatiation:** every neighborhood of a consumption bundle contains a strictly preferred bundle.
- **Complete markets:** all relevant goods, delivery dates and states can be traded.
- **No externalities:** effects on others operate through market prices and trades.

Convexity is not required. Economies with *infinitely many overlapping generations* lie outside this finite-consumer result; competitive equilibria can be inefficient.

2.10.2. Proof by contradiction of Theorem 2.10.1

Proof: Let $p \geq 0$ denote equilibrium prices and ω aggregate endowments. At equilibrium, consumer i consumes x_i^* and has wealth m_i from endowments and distributed profits; firm j produces y_j^* .

Suppose $(\tilde{x}_i, \tilde{y}_j)$ is a feasible Pareto improvement: every consumer weakly prefers $\tilde{x}_i$ to x_i^* , and consumer h strictly prefers $\tilde{x}_h$ to x_h^* .

1. Consumer optimality.

- **Strict improvement:** the bundle must cost more than m_i ; otherwise it was affordable.
- **Weak improvement:** the bundle cannot cost less than m_i . With *budget slack*, local nonsatiation gives a nearby, strictly better bundle that remains affordable.

Either violation contradicts optimality. Hence

$$\begin{aligned} p \cdot \tilde{x}_i &\geq m_i \text{ for every } i, \\ p \cdot \tilde{x}_h &> m_h \text{ for the strictly improved consumer } h. \end{aligned} \quad (2.52)$$

Summing gives

$$\sum_i p \cdot \tilde{x}_i > \sum_i m_i. \quad (2.53)$$

2. **Feasibility and firm optimality.** Consumption is bounded by endowments plus production.

Profit maximization implies $p \cdot \tilde{y}_j \leq p \cdot y_j^*$ for each firm. Therefore

$$\begin{aligned} \sum_i p \cdot \tilde{x}_i &\leq p \cdot \omega + \sum_j p \cdot \tilde{y}_j \text{ (feasibility),} \\ &\leq p \cdot \omega + \sum_j p \cdot y_j^* \text{ (profit maximization),} \\ &= \sum_i m_i \text{ (endowments + distributed profits).} \end{aligned} \tag{2.54}$$

3. **Contradiction.** Total cost cannot be both *greater than* and *at most* aggregate wealth. No feasible Pareto improvement exists.

□

2.10.3. Implication for the standard model

In the **representative-agent standard model**, a competitive equilibrium allocation solves the planner's problem. *If the planner's solution is unique and equilibrium exists*, the allocations coincide. Recover prices from firms' marginal products.

2.11. The Second Welfare Theorem

Theorem 2.11.1 (Second Welfare Theorem): With complete competitive markets, no externalities and the conditions below, any **Pareto-efficient allocation** can be supported as a **competitive equilibrium** after **lump-sum wealth transfers**.

2.11.1. Sufficient conditions

- **Consumers:** a finite set.
- **Consumption:** each consumption set X_i is convex.
- **Preferences:** u_i is continuous and locally nonsatiated. For $x, x' \in X_i$ and $0 < \lambda < 1$, assume

$$u_i(x) > u_i(x') \implies u_i(\lambda x + (1 - \lambda)x') > u_i(x'). \tag{2.55}$$

- **Production:** the aggregate production set $Y = \sum_j Y_j$ is convex.
- **Separation:** the commodity space S is finite-dimensional, or Y has a nonempty interior.
- **Budget regularity:** at supporting prices p , each consumer has a feasible bundle $x_i^0 \in X_i$ with

$$p \cdot x_i^0 < p \cdot x_i^*. \tag{2.56}$$

Convexity alone guarantees only a *quasi-equilibrium*: a strictly preferred bundle may cost exactly the consumer's wealth.¹

¹See [Condie's welfare-theorem notes, Proposition 6.1](#) ° for the budget condition. In an infinite-dimensional commodity space, $p \cdot x$ denotes the continuous price functional applied to x .

2.11.2. Wealth transfers

At supporting prices, m_i includes original endowments and profits at the target production plan.

For target consumption x_i^* , choose **transfers** T_i :

$$\begin{aligned} T_i &= \underbrace{p \cdot x_i^*}_{\text{target wealth}} - \underbrace{m_i}_{\text{original wealth}}, \\ p \cdot x_i &\leq m_i + T_i = p \cdot x_i^*, \\ \sum_i T_i &= 0. \end{aligned} \tag{2.57}$$

Feasibility and profit distribution imply $\sum_i p \cdot x_i^* = \sum_i m_i$, so transfers sum to zero. **Lump-sum** transfers are *independent of choices*, leaving marginal incentives unchanged.

2.11.3. Supporting prices

1. **Separation:** convexity permits a separating hyperplane between feasible aggregate resources and aggregate bundles that improve welfare.
2. **Prices:** the separating functional defines the price system p .
3. **Optimization:** firms maximize profits at these prices; budget regularity ensures consumers optimize their transfer-adjusted budgets.

2.12. How do the welfare theorems matter?

- **Markets and information:** under the welfare-theorem assumptions, equilibrium prices coordinate decentralized choices into an efficient allocation. *The Use of Knowledge in Society* (Hayek, 1945) explains how prices communicate *dispersed information*.
- **Policy:** diagnose failed assumptions, such as externalities, market power or missing markets. Under the second theorem's conditions, *lump-sum redistribution* can implement different efficient distributions without changing marginal incentives.
- **Computing equilibrium:** in the standard model, solve the planner's Bellman equation, then recover prices from marginal products. With heterogeneity, a planner allocation may require *wealth transfers* to be decentralized.

3. Lecture 4 Oct 6

3.1. Fundamental models of economic growth

The **standard model** is a Solow growth model with *optimizing consumers*. It matches long-run growth facts for high-income countries, but growth models should also explain:

- Economic development beyond the period described by the Kaldor facts.
- Income and growth differences across *all countries*.
- The origin of productivity growth.

Definition 3.1.1 (Kaldor's growth facts): Kaldor's six **stylized facts** (Kaldor, 1961) summarize long-run growth patterns, originally observed in industrialized economies:

1. **Output per worker** $\frac{Y_t}{N_t}$ grows at a roughly constant positive rate.
2. **Capital per worker** $\frac{K_t}{N_t}$ grows over time.
3. The **return to capital** is roughly constant.
4. The **capital-output ratio** $\frac{K_t}{Y_t}$ is roughly constant.
5. **Labor and capital income shares** are roughly constant.
6. **Output-per-worker growth rates** differ substantially across countries.

3.2. Solow

3.2.1. Setup

Output combines capital K_t and effective labor $A_t N_t$:

$$Y_t = K_t^\alpha (A_t N_t)^{1-\alpha}, 0 < \alpha < 1. \quad (3.1)$$

Here N_t is population (and labor supply); A_t is **labor-augmenting productivity**. Both grow *exogenously*:

$$\begin{aligned} A_{t+1} &= (1 + g)A_t, \\ N_{t+1} &= (1 + n)N_t. \end{aligned} \quad (3.2)$$

Assume $g, n \geq 0$. A **fixed saving rate** $0 < s < 1$ determines investment:

$$\begin{aligned} K_{t+1} &= (1 - \delta)K_t + sY_t, \\ C_t &= (1 - s)Y_t, \end{aligned} \quad (3.3)$$

where $0 \leq \delta \leq 1$ is depreciation. In the *standard model*, households instead choose saving optimally.

3.2.2. Capital per effective worker

Normalize by effective labor:

$$\begin{aligned}k_t &= \frac{K_t}{A_t N_t}, \\y_t &= \frac{Y_t}{A_t N_t} = k_t^\alpha, \\c_t &= \frac{C_t}{A_t N_t} = (1 - s)k_t^\alpha.\end{aligned}\tag{3.4}$$

Divide capital accumulation by $A_t N_t$ and use $A_{t+1} N_{t+1} = (1 + g)(1 + n)A_t N_t$:

$$\begin{aligned}k_{t+1}(1 + g)(1 + n) &= (1 - \delta)k_t + sk_t^\alpha, \\k_{t+1} - k_t &= \frac{sk_t^\alpha - (\delta + g + n + gn)k_t}{(1 + g)(1 + n)}.\end{aligned}\tag{3.5}$$

Effective capital rises when investment exceeds depreciation and the capital needed to keep up with population and productivity growth. The term gn is retained because time is *discrete*.

3.2.3. Balanced growth path

On a **balanced growth path (BGP)**, capital per effective worker is constant:

$$\begin{aligned}s\bar{k}^\alpha &= (\delta + g + n + gn)\bar{k}, \\ \bar{k} &= \left(\frac{s}{\delta + g + n + gn} \right)^{\frac{1}{1-\alpha}}.\end{aligned}\tag{3.6}$$

This positive steady state requires $\delta + g + n + gn > 0$. For $k_0 > 0$, diminishing returns imply convergence: k_t rises below $\bar{k}$ and falls above it.

- **Aggregate quantities:** K_t , Y_t and C_t grow at rate $(1 + g)(1 + n) - 1$.
- **Per capita quantities:** $\frac{K_t}{N_t}$, $\frac{Y_t}{N_t}$ and $\frac{C_t}{N_t}$ grow at rate g .
- **Growth facts:** the capital-output ratio, factor income shares and the marginal product of capital are constant; real wages grow at rate g .

A higher saving rate raises the BGP **level** of income per capita, not its **growth rate** g . With $g > 0$, the BGP matches Kaldor's first five facts. Cross-country differences in long-run growth require different *exogenous* productivity growth rates g .

3.2.4. Limitations

- Persistent cross-country income gaps require *exogenous differences* in productivity, saving or population growth.
- Baseline calibrations can imply faster convergence than observed, even among high-income countries.
- Productivity growth is unexplained; the model also lacks a mechanism for the historical transition from long-run stagnation to sustained growth.

3.3. Malthus

3.3.1. Setup

The model describes a preindustrial economy with population N_t and **fixed land** $X > 0$:

$$Y_t = N_t^\alpha (A_t X)^{1-\alpha}, 0 < \alpha < 1. \quad (3.7)$$

Productivity grows exogenously at rate $g \geq 0$:

$$A_{t+1} = (1 + g)A_t. \quad (3.8)$$

Population growth responds to income per capita:

$$\begin{aligned} \frac{N_{t+1}}{N_t} &= s \frac{Y_t}{N_t}, \\ N_{t+1} &= sY_t. \end{aligned} \quad (3.9)$$

Here $s > 0$ is a demographic parameter. Higher income raises fertility or reduces mortality. Population increases when $s \frac{Y_t}{N_t} > 1$.

3.3.2. Population per effective land unit

Normalize by effective land $A_t X$:

$$\begin{aligned} n_t &= \frac{N_t}{A_t X}, \\ \frac{Y_t}{A_t X} &= n_t^\alpha, \\ \frac{Y_t}{N_t} &= n_t^{\alpha-1}. \end{aligned} \quad (3.10)$$

Using $N_{t+1} = sY_t$ gives

$$\begin{aligned} n_{t+1}(1 + g) &= s n_t^\alpha, \\ n_{t+1} &= \frac{s}{1 + g} n_t^\alpha. \end{aligned} \quad (3.11)$$

Since $\alpha < 1$, income per person falls as population grows relative to effective land.

3.3.3. Balanced growth path

On the BGP, $n_t = \bar{n}$ is constant:

$$\begin{aligned} \bar{n}(1 + g) &= s \bar{n}^\alpha, \\ \bar{n} &= \left(\frac{s}{1 + g} \right)^{\frac{1}{1-\alpha}}. \end{aligned} \quad (3.12)$$

For $n_0 > 0$, n_t converges to $\bar{n}$: it rises below the steady state and falls above it.

BGP income per capita and population density are

$$\begin{aligned}\frac{Y_t}{N_t} &= \bar{n}^{\alpha-1} = \frac{1+g}{s}, \\ \frac{N_t}{X} &= A_t \bar{n}.\end{aligned}\tag{3.13}$$

- Population and aggregate output grow at rate g .
- **Income per capita is constant, even with positive productivity growth.**
- Holding s and g fixed, higher productivity levels imply greater population density and the same BGP income per capita.

A permanent increase in the *level* of productivity initially raises income per person. Higher income accelerates population growth. As population rises relative to land, diminishing returns bring income per person back toward $\frac{1+g}{s}$.

- Explains the world until 1800s

3.3.4. Malthus versus Solow

Solow accumulates **capital**; Malthus accumulates **population**. On the BGP, the return to the accumulated factor is constant. With competitive factor pricing, Malthus implies

$$\begin{aligned}w_t &= \alpha \frac{Y_t}{N_t} = \alpha \frac{1+g}{s}, \\ r_t^X &= (1-\alpha) \frac{Y_t}{X},\end{aligned}\tag{3.14}$$

where r_t^X is the rental rate of land. *Wages are constant and land rents grow at rate g .*

In Solow, the return to capital is constant and wages grow at rate g .

3.3.5. From Malthus to Solow

Malthus describes preindustrial stagnation; Solow describes sustained growth in modern economies. Neither model alone explains the transition. *Malthus to Solow* (Hansen & Prescott, 2002) combines both technologies in one economy. Firms choose which to operate as productivity grows.

3.3.5.1. Technologies and resources

Both technologies are available in every period and produce the **same good**:

$$\begin{aligned}Y_{M,t} &= A_{M,t} K_{M,t}^{\alpha_1} N_{M,t}^{\alpha_2} X^{1-\alpha_1-\alpha_2}, \\ Y_{S,t} &= A_{S,t} K_{S,t}^{\theta} N_{S,t}^{1-\theta}.\end{aligned}\tag{3.15}$$

The Malthus sector M uses land; the Solow sector S uses only capital and labor. Assume $\alpha_1, \alpha_2 > 0$, $\alpha_1 + \alpha_2 < 1$ and $0 < \theta < 1$. Sectoral **TFP** grows exogenously:

$$A_{j,t+1} = \gamma_j A_{j,t}, \quad j \in \{M, S\}.\tag{3.16}$$

Here $\gamma_M \geq 1$ and $\gamma_S > 1$ are gross growth factors. Competitive firms allocate capital and labor freely across sectors:

$$\begin{aligned} K_t &= K_{M,t} + K_{S,t}, \\ N_t &= N_{M,t} + N_{S,t}, \\ X &= 1. \end{aligned} \tag{3.17}$$

One period represents a generation, so capital depreciates fully ($\delta = 1$). Land does not depreciate. Aggregate feasibility is

$$C_t + K_{t+1} = Y_{M,t} + Y_{S,t}. \tag{3.18}$$

3.3.5.2. Households and population

Households live for two periods and work only when young. A household born at t chooses positive consumption and nonnegative capital k_{t+1} and land x_{t+1} . With $\beta > 0$, its objective is

$$\max \log c_{1,t} + \beta \log c_{2,t+1}, \tag{3.19}$$

subject to

$$\begin{aligned} c_{1,t} + k_{t+1} + q_t x_{t+1} &= w_t, \\ c_{2,t+1} &= r_{t+1}^K k_{t+1} + (r_{t+1}^X + q_{t+1}) x_{t+1}. \end{aligned} \tag{3.20}$$

Here q_t is the land price and r_t^K, r_t^X are capital and land rental rates. With full depreciation, r_{t+1}^K is the entire payoff from one unit of capital saved at t .

There are N_t young households at t . Population follows a **calibrated** function G :

$$N_{t+1} = G(c_{1,t}) N_t. \tag{3.21}$$

G rises with consumption at low living standards and falls at high living standards. Each young household supplies one unit of labor. Asset and goods markets clear when

$$\begin{aligned} K_{t+1} &= N_t k_{t+1}, \\ 1 &= N_t x_{t+1}, \\ C_t &= N_t c_{1,t} + N_{t-1} c_{2,t}. \end{aligned} \tag{3.22}$$

With interior asset holdings, household optimality gives

$$\begin{aligned} c_{1,t} &= \frac{w_t}{1 + \beta}, \\ q_t r_{t+1}^K &= r_{t+1}^X + q_{t+1}, \\ K_{t+1} &= N_t \frac{\beta}{1 + \beta} w_t - q_t. \end{aligned} \tag{3.23}$$

The second equation equates the returns on capital and land. The third subtracts land purchases from the young generation's total saving.

3.3.5.3. Solow sector entry

Normalize the output price to one. At given factor prices, the minimum cost of producing one unit with the Solow technology is

$$\frac{1}{A_{S,t}} \left(\frac{r_t^K}{\theta} \right)^\theta \left(\frac{w_t}{1-\theta} \right)^{1-\theta}. \quad (3.24)$$

To test entry, evaluate this cost at the **Malthus-only** factor prices $r_{M,t}^K$ and $w_{M,t}$. Entry is profitable when

$$A_{S,t} > \underbrace{\left(\frac{r_{M,t}^K}{\theta} \right)^\theta \left(\frac{w_{M,t}}{1-\theta} \right)^{1-\theta}}_{\text{entry threshold}}. \quad (3.25)$$

Once entry occurs, factor prices adjust. If both sectors operate, their marginal products of capital and labor must coincide:

$$\begin{aligned} r_t^K &= \alpha_1 \frac{Y_{M,t}}{K_{M,t}} = \theta \frac{Y_{S,t}}{K_{S,t}}, \\ w_t &= \alpha_2 \frac{Y_{M,t}}{N_{M,t}} = (1-\theta) \frac{Y_{S,t}}{N_{S,t}}. \end{aligned} \quad (3.26)$$

The Malthus sector remains active at every finite date. With fixed land, scaling its capital and labor down together makes their marginal products arbitrarily large, so allocating a small amount to this sector remains worthwhile.

3.3.5.4. Transition

1. Initially, only the Malthus technology operates. Population growth absorbs productivity gains, keeping living standards constant. Solow productivity continues to grow while that sector is unused.
2. On the Malthusian growth path, wages and capital rental rates are constant, so the entry threshold is fixed. Rising $A_{S,t}$ eventually crosses it, and firms allocate capital and labor to the Solow sector.
3. The Solow sector has constant returns in capital and labor, with no fixed land input. In the calibrated economy, its share expands and income per person grows. The Malthus sector's share tends to zero. Population growth eventually falls through the assumed function G .

3.3.5.5. Limitations

- Productivity growth remains exogenous.
- The demographic transition comes from the calibrated function G ; fertility and mortality decisions are not derived from household choices.
- The two technologies produce the same good. Requiring a separate agricultural good for subsistence would change the mechanism.

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